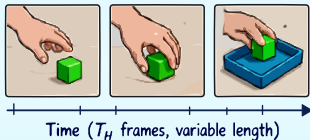


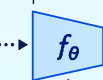
(a) Shared representations

Human video (place cube into tray)



Cached interaction tokens

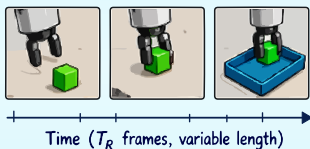
512-D per token
variable token count per frame



128-D
frame embeddings

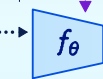
Shared weights

Robot video (place cube into tray)



Cached interaction tokens

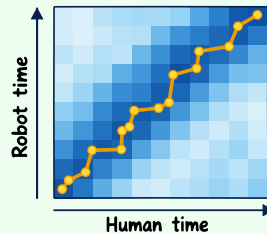
512-D per token
variable token count per frame



128-D
frame embeddings

(b) Monotonic matching

Cosine similarity matrix (schematic)



DTW path (monotonic)

Two-way DTW

Robot-to-human
matches

Human-to-robot
matches

(c) Filtered correspondences

Filtering criteria



Similarity

Cycle consistency

Match margin

All pass

Retained frame pairs

Human

Robot



✓



✓

Rejected (e.g. low similarity or inconsistent)



✗

(d) Shared adapter (frozen at inference)

